

MUSTAFA AHMED

AI Agent Engineer | Backend Systems & Production AI Deployment

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SUMMARY

AI agent engineer with a backend-systems foundation, focused on designing and hardening AI systems for production: permission and guardrail architecture, prompt-injection defense, and evaluation, built hands-on in Rust, Python and TypeScript. Currently architecting Standwise, a multi-tenant, chat-first work platform with AI standups powered by Claude, grounded outputs and CI-gated evals. Previously shipped RESTful APIs at sub-200ms latency for a tour-booking platform.

SKILLS

Agent Engineering: Multi-agent orchestration & system design, permission/guardrail architecture, risk-tiered access control, human-in-the-loop approval

AI/ML Tooling: Anthropic Claude API, LangChain, LangGraph, PaddleOCR, RAG, pgvector, structured outputs, agent evaluation, prompt-injection testing

Languages: Rust, Python, Go, TypeScript, JavaScript

Backend & Data: Node.js, NestJS, Express, chi, fiber, sqlx, Prisma, PostgreSQL (RLS), Redis, SQL, SQLite

Cloud/Infra: AWS (S3, SQS, EKS, ECS, SES, RDS), Docker, Kubernetes, Terraform, Linux, Git

Frontend: React.js, React Native

AI AGENT ENGINEERING PROJECTS

Multi-Agent DevOps Incident Response System — Rust, PostgreSQL

- Hand-built a 3-agent system (Diagnostic, Remediation, Communication) with isolated context per agent, no orchestration framework, and risk-tiered permissions across 4 infrastructure action types
- Found and closed a real cross-service permission gap via a targeted prompt-injection test using poisoned runbook content, then verified the fix with regression tests
- Built an automated eval suite with a held-out test set and phase-tagged tracing across every agent handoff

Order Support Agent — Python, LangGraph

- Built a LangGraph-orchestrated agent with a 3-tier permission system (auto-approve, log-only, human-confirm) that escalates a real fraud pattern: shipping-address changes on already-shipped orders
- Proved permission enforcement holds under adversarial pressure, testing direct pushback and injection-style bypass attempts at both the action and scope level
- Found and fixed a metrics-scoping bug and a dead-code gap where a safety fix wasn't wired into the live serving path; validated with a 6-case automated eval suite

WORK EXPERIENCE

ByteForge — Solution Engineer

Dec 2025 – Present

- Architected Standwise, a multi-tenant, chat-first work platform (HR + project-management modules) designed as a modular TypeScript monolith (NestJS, PostgreSQL, Redis, AWS ECS Fargate), with ByteForge as tenant #1
- Designed an AI standup engine on Claude, routing Haiku, Sonnet and Opus by task, that parses free-text updates into structured fields, detects implied blockers, and publishes summaries and lead alerts to Slack
- Applied a "code decides, LLM writes" architecture: sprint detection, risk rules, metrics and permissions run in tested code, so the model only writes language and cannot invent numbers
- Designed LLM guardrails: zod-validated structured outputs with retry, versioned prompts, redaction of sensitive HR data from model context, human approval before AI writes to Jira or calendars, audit logging of every agent action, and CI evals with number-grounding hallucination checks that block merges on regression
- Designed three-layer tenant isolation (application-level query scoping, PostgreSQL row-level security, CI isolation test suite) plus scoped RBAC and entitlement-gated features

Khel — Co-Founder & Backend Engineer (part-time, concurrent with ByteForge)

Jan 2026 – Present

- Co-founded Khel and own technical vision across backend and mobile; architected the reservation system, resolving booking conflicts, payment deadlines, and partial payments to directly shape booking-to-revenue conversion
- Designed a peer-to-peer split-payment system, replacing third-party gateway integration to cut transaction costs and external dependency
- Built an AI-assisted ELO rating system that tracks player performance across matches to power skill-based matchmaking, plus gamification (badges, streaks) to drive user retention

Infoetech — Software Engineer

Nov 2023 – Dec 2025

- Designed and delivered RESTful APIs and microservices (Node.js, Express, Socket.IO) powering search, availability, and checkout at sub-200ms response times; optimized performance via index tuning and Redis caching, and directed zero-downtime AWS deployments (Docker, Kubernetes)

Ethisol — Software Engineer

Mar 2023 – Nov 2023

- Designed scalable backend architecture (Node.js, MongoDB) with Stripe/PayPal integration, RBAC, and JWT-based authentication meeting enterprise security requirements

EDUCATION

University of Lahore — Bachelor of Computer Science